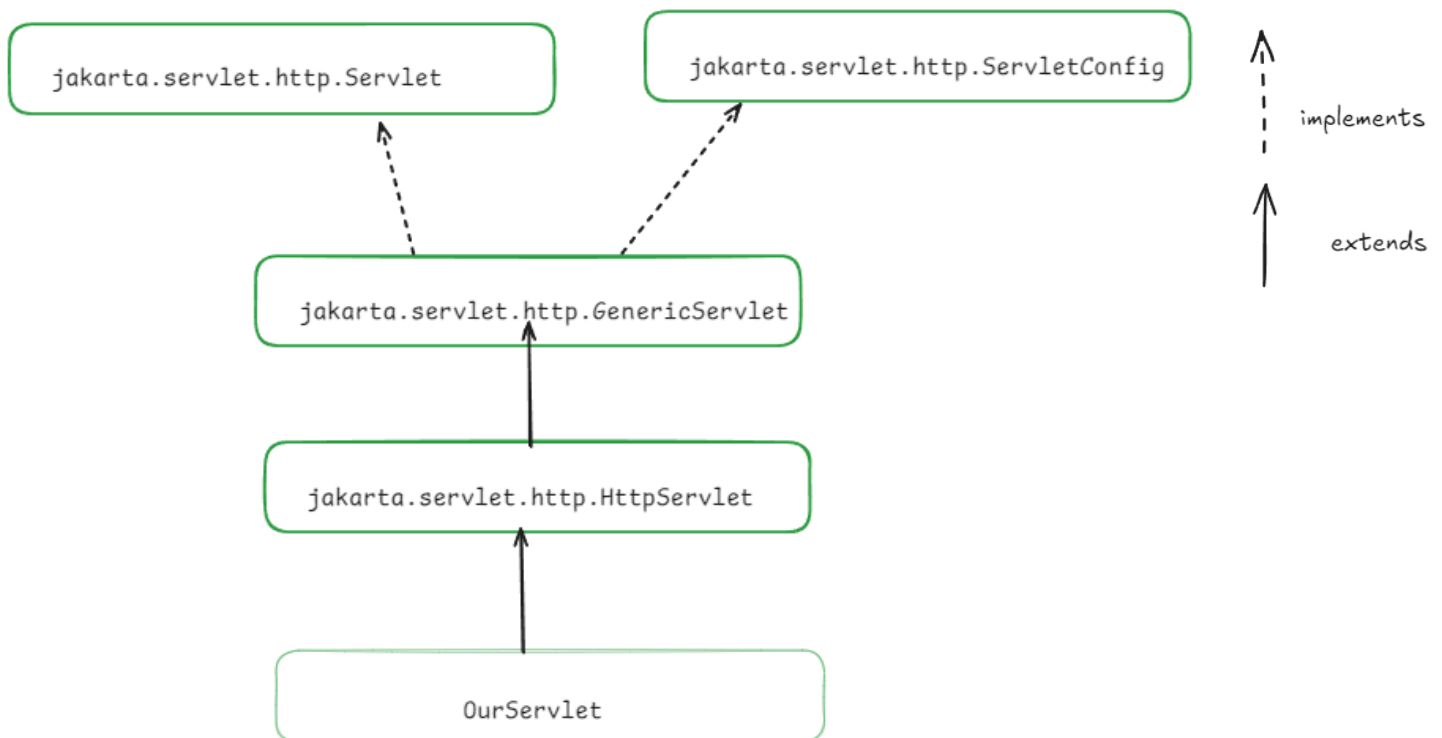


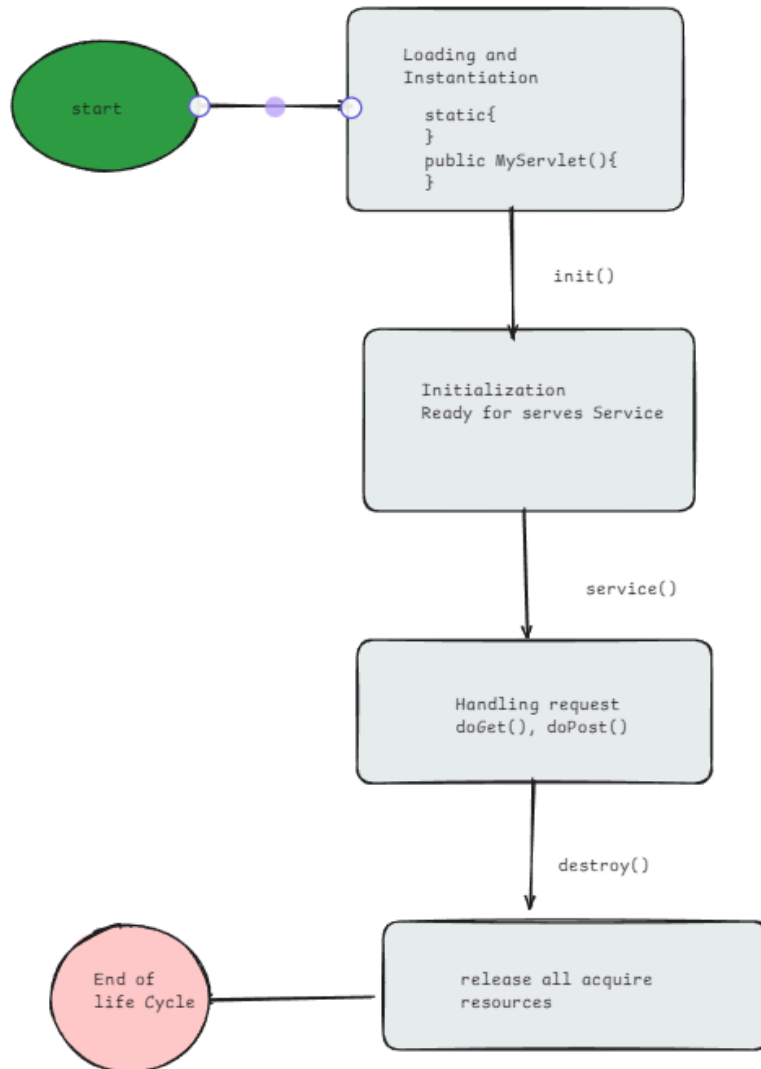
Servlet Lifecycle

The servlet lifecycle is the process through which a servlet goes from creation to destruction in a web application. Understanding this lifecycle is crucial for effective servlet management. Here are the key stages:

👉 Hierarchy of Servlet



👉 Stages of Servlet lifecycle



1. Loading and Instantiation

- **Loading:** The servlet container (e.g., Tomcat) loads the servlet class when it is first requested or at server startup if it is configured for loading at startup.
- **Instantiation:** The container creates an instance of the servlet class using the default constructor.

2. Initialization (init Method)

- **Purpose:** The init() method is called by the servlet container after instantiation and is used for initializing the servlet.
- **Signature:** public void init(ServletConfig config) throws ServletException
- **Initialization Parameters:** You can access initialization parameters via the ServletConfig object passed to the init() method.
- **Note:** The init() method is called only once during the lifecycle of the servlet.

3. Request Handling (service Method)

- **Purpose:** The service() method is called by the container each time a request is made to the servlet.
- **Signature:** public void service(ServletRequest req, ServletResponse res) throws ServletException, IOException
- **Process:**
 - The service() method determines the type of request (e.g., GET, POST) and dispatches it to the appropriate handler method (doGet(), doPost(), etc.).
 - **Multithreading:** The service() method can be called concurrently for multiple requests, so servlets must be thread-safe.
- **Common Methods:**

- doGet(HttpServletRequest req, HttpServletResponse res)
- doPost(HttpServletRequest req, HttpServletResponse res)

4. Destruction (destroy Method)

- **Purpose:** The destroy() method is called by the servlet container to indicate that the servlet is being taken out of service.
- **Signature:** public void destroy()
- **Clean-Up:** This method is used to release any resources (e.g., database connections, file handles) that the servlet might be holding.
- **Note:** The destroy() method is called only once at the end of the servlet's lifecycle, just before the servlet object is garbage-collected.

Additional Points:

- **Single Instance:** Typically, only one instance of the servlet is created, regardless of the number of requests.
- **Thread Safety:** Since multiple threads may access the service() method simultaneously, developers must ensure that the servlet code is thread-safe.
- **ServletConfig and ServletContext:**
 - **ServletConfig:** Provides configuration information to the servlet.
 - **ServletContext:** Allows the servlet to interact with its environment (e.g., to access application-wide parameters).